Business Case: Delhivery - Feature Engineering

1.Problem Statement

India's leading logistics and supply chain companies that manage shipments and deliveries across multiple transportation routes and operational centers.

The logistics industry generates massive amounts of operational data related to:
- Delivery performance
- Shipment timings
- Transportation routes
- Distance calculations
- Warehouse coordination
- Operational efficiency

Analyzing logistics data is important because it helps companies:
- Improve delivery speed
- Reduce transportation delays
- Optimize operational performance
- Increase customer satisfaction
- Reduce logistics costs

The objective of this project is to perform detailed Root Cause Analysis (RCA) and Feature Engineering on Delhivery’s logistics dataset using Python and Data Analytics techniques.

This project focuses on:
1. Understanding the dataset structure

2. Cleaning and preprocessing logistics data
3. Handling missing values
4. Performing Exploratory Data Analysis (EDA)
5. Creating meaningful features
6. Aggregating trip-level operational data
7. Performing hypothesis testing
8. Detecting outliers
9. Generating business insights
10. Providing recommendations for operational improvement

2. Introduction

It's one of the biggest logistics and supply chain companies in India. It handles deliveries across different transportation routes, warehouses, and operational centers.

In the logistics industry, a large amount of operational data is generated every day related to delivery performance, shipment timings, route distances, warehouse management, and transportation efficiency.

Analyzing logistics data is very important because it helps companies to:
- Improve delivery speed
- Reduce transportation delays
- Increase operational efficiency
- Improve customer satisfaction
- Reduce overall logistics costs

The main objective of this project is to perform Root Cause Analysis (RCA) and Feature Engineering on Delhivery's logistics dataset using Python and Data Analytics techniques.

In this project, we will:
1. Understand the structure of the dataset
2. Clean and preprocess the logistics data
3. Handle missing values
4. Perform Exploratory Data Analysis (EDA)
5. Create meaningful features from the dataset
6. Aggregate trip-level operational data
7. Perform hypothesis testing
8. Detect outliers in the data
9. Generate business insights
10. Provide recommendations to improve operational performance

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats import ttest_ind

from sklearn.preprocessing import StandardScaler

import warnings
warnings.filterwarnings('ignore')
```

3.Dataset Loading & Initial Exploration

The dataset was loaded successfully into the Python environment using the Pandas library. After loading the dataset, the first five rows were displayed using the `head()` function to get an initial understanding of the data.

Initial exploration of the dataset is very important because it helps in understanding:
- The overall structure of the dataset
- Different column names and their meanings

- Logistics and transportation-related variables
- Operational metrics present in the dataset
- Data types and possible missing values

The dataset contains several important logistics-related features such as:
- Trip creation time
- Route type
- Source and destination centers
- Actual delivery time
- Estimated delivery time
- Actual transportation distance
- Estimated transportation distance
- Segment-level operational details

By performing initial exploration, it becomes easier to understand the complexity of the logistics operations and prepare the dataset for further preprocessing and analysis. This step also helps in identifying missing values, incorrect data types, duplicate records, and other issues before moving toward detailed Exploratory Data Analysis (EDA), Feature Engineering, and Hypothesis Testing.

```
df.head()
```

	data	trip_creation_time	route_schedule_uuid	route_type	trip_uuid	source_center	source_name	destination_center	destination_
0	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDF (Gu
1	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDF (Gu
2	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDF (Gu
3	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDF (Gu
4	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDF (Gu

5 rows × 24 columns

4.Dataset Shape Analysis

The `shape()` function was used to check the total number of rows and columns present in the dataset. This helps in understanding how large the dataset is before starting detailed analysis.

Purpose of Shape Analysis

Shape analysis is important because it gives basic information about:
- Total number of observations
- Total number of variables
- Size of the dataset
- Complexity of the analysis process

Observation

After checking the shape of the dataset, it was observed that the dataset contains a large number of records along with multiple logistics and operational variables related to delivery performance and transportation activities.

Interpretation

A large dataset is useful because it provides better analytical results and more accurate insights. It also helps in identifying patterns, operational issues, and trends more effectively during analysis.

Business Importance

Understanding the size of the dataset is very important in logistics analytics because large operational datasets help companies analyze delivery performance, transportation efficiency, and operational bottlenecks in a better way. It also improves decision-making and helps in generating more reliable business insights.

```
df.shape
```

```
(144867, 24)
```

5. Dataset Information & Data Types

The `info()` function was used to understand the overall structure of the dataset and check the datatype of each column present in the logistics dataset.

Purpose of Data Type Analysis

Analyzing data types is important because it helps in:
- Identifying numerical columns
- Identifying categorical columns
- Understanding datetime-related variables
- Detecting missing values
- Preparing the dataset for preprocessing and analysis

Observation

From the dataset information, it was observed that the dataset contains different types of variables such as:
- Integer variables
- Float variables
- Object variables
- Datetime-related variables

The dataset also contains important operational features related to logistics such as:
- Delivery timing information
- Distance calculations
- Route details
- Transportation-related operational metrics

Interpretation

Different datatypes require different preprocessing methods during analysis. Numerical variables may require scaling, normalization, and outlier treatment, while categorical variables may require encoding techniques for machine learning and analytical purposes.

Datetime variables are also important because they help in extracting useful time-based insights such as trip duration, delay analysis, and delivery patterns.

Business Importance

Understanding the datatype of each variable is very important in logistics analytics because it helps in selecting suitable

preprocessing and analytical techniques. Proper datatype analysis improves data quality, helps in accurate feature engineering, and supports better operational decision-making.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144867 entries, 0 to 144866
Data columns (total 24 columns):
 #   Column                          Non-Null Count   Dtype  
---  ------                          --------------   -----  
 0   data                            144867 non-null  object 
 1   trip_creation_time              144867 non-null  object 
 2   route_schedule_uuid             144867 non-null  object 
 3   route_type                      144867 non-null  object 
 4   trip_uuid                       144867 non-null  object 
 5   source_center                   144867 non-null  object 
 6   source_name                     144574 non-null  object 
 7   destination_center              144867 non-null  object 
 8   destination_name                144606 non-null  object 
 9   od_start_time                   144867 non-null  object 
 10  od_end_time                     144867 non-null  object 
 11  start_scan_to_end_scan          144867 non-null  float64
 12  is_cutoff                       144867 non-null  bool   
 13  cutoff_factor                   144867 non-null  int64  
 14  cutoff_timestamp                144867 non-null  object 
 15  actual_distance_to_destination  144867 non-null  float64
 16  actual_time                     144867 non-null  float64
 17  osrm_time                       144867 non-null  float64
 18  osrm_distance                   144867 non-null  float64
 19  factor                          144867 non-null  float64
 20  segment_actual_time             144867 non-null  float64
 21  segment_osrm_time               144867 non-null  float64
 22  segment_osrm_distance           144867 non-null  float64
 23  segment_factor                  144867 non-null  float64
dtypes: bool(1), float64(10), int64(1), object(12)
memory usage: 25.6+ MB
```

6.Statistical Summary of Dataset

The `describe()` function was used to generate the statistical summary of the numerical variables present in the logistics dataset. This helped in understanding the overall distribution and behavior of the operational data.

Purpose of Statistical Analysis

Statistical analysis is useful because it helps in:
- Understanding the average values of variables
- Understanding data distribution
- Measuring spread and variability
- Detecting extreme values
- Identifying possible outliers in the dataset

Observation

From the statistical summary, it was observed that there are large variations in several logistics-related variables such as:
- Delivery time
- Transportation distance
- Segment-level operational metrics

Some variables show a huge difference between minimum and maximum values, which indicates the presence of operational inconsistencies and possible outliers in the dataset.

Interpretation

The high variation in logistics data may occur due to several operational reasons such as:
- Traffic congestion
- Route inefficiencies
- Delivery delays
- Warehouse processing delays
- Long-distance transportation routes

These factors can significantly affect delivery performance and operational efficiency.

Business Importance

Understanding the statistical distribution of logistics variables is very important because it helps companies identify inefficient transportation routes, operational bottlenecks, unusual delivery behavior, and performance issues. This analysis supports better operational planning and improves overall logistics efficiency.

```
df.describe()
```

	start_scan_to_end_scan	cutoff_factor	actual_distance_to_destination	actual_time	osrm_time	osrm_distance	factor	segment_actual_time	segment_osrm_tim
count	144867.000000	144867.000000	144867.000000	144867.000000	144867.000000	144867.000000	144867.000000	144867.000000	144867.00000
mean	961.262986	232.926567	234.073372	416.927527	213.868272	284.771297	2.120107	36.196111	18.50754
std	1037.012769	344.755577	344.990009	598.103621	308.011085	421.119294	1.715421	53.571158	14.77596
min	20.000000	9.000000	9.000045	9.000000	6.000000	9.008200	0.144000	-244.000000	0.00000
25%	161.000000	22.000000	23.355874	51.000000	27.000000	29.914700	1.604264	20.000000	11.00000
50%	449.000000	66.000000	66.126571	132.000000	64.000000	78.525800	1.857143	29.000000	17.00000
75%	1634.000000	286.000000	286.708875	513.000000	257.000000	343.193250	2.213483	40.000000	22.00000
max	7898.000000	1927.000000	1927.447705	4532.000000	1686.000000	2326.199100	77.387097	3051.000000	1611.00000

7. Missing Value Analysis

Missing value analysis was performed to check whether the dataset contains any incomplete or null records in the logistics data.

Purpose of Missing Value Analysis

Analyzing missing values is important because it helps in:
- Identifying incomplete data
- Understanding overall data quality
- Improving preprocessing accuracy
- Avoiding analytical errors
- Improving the reliability of analysis and machine learning models

Observation

After checking the dataset, it was observed that only a few missing values are present across some operational variables.

Most of the columns are complete and suitable for further logistics analysis and feature engineering.

Interpretation

Missing values can affect:
- Statistical analysis
- Data visualization
- Machine learning model performance
- Business insights and interpretation

Therefore, handling missing values properly is necessary before moving toward advanced analysis, feature engineering, and hypothesis testing.

Business Importance

Proper handling of missing values improves the overall quality of analysis and helps generate more accurate operational and transportation insights. It also helps companies make better business decisions based on reliable logistics data.

```
df.isnull().sum()
```

	0
data	0
trip_creation_time	0
route_schedule_uuid	0
route_type	0
trip_uuid	0
source_center	0
source_name	31
destination_center	0
destination_name	14
od_start_time	1
od_end_time	1
start_scan_to_end_scan	1
is_cutoff	1

8. Missing Value Visualization

A heatmap was created to visually analyze the missing values present in the logistics dataset. The visualization helped in understanding how missing data is distributed across different variables in the dataset.

Missing value visualization is an important step during data preprocessing because it gives a clear overview of dataset quality and helps identify whether any column contains a large amount of incomplete information.

Purpose of Visualization

Visualizing missing values is useful because it helps in:
- Understanding patterns of missing data
- Detecting columns with incomplete information
- Identifying sparse variables
- Improving preprocessing decisions
- Understanding overall dataset quality
- Preparing the dataset for accurate analysis and modeling

Observation

From the heatmap, it was observed that most of the variables contain very few missing values. This indicates that the dataset has good overall data quality and is suitable for detailed logistics analysis.

Only a small number of variables contain limited missing information, and the missing data is not heavily concentrated in any particular section of the dataset.

Interpretation

Datasets with fewer missing values are generally easier to preprocess and analyze. They also produce more reliable statistical results, visualizations, and machine learning outputs.

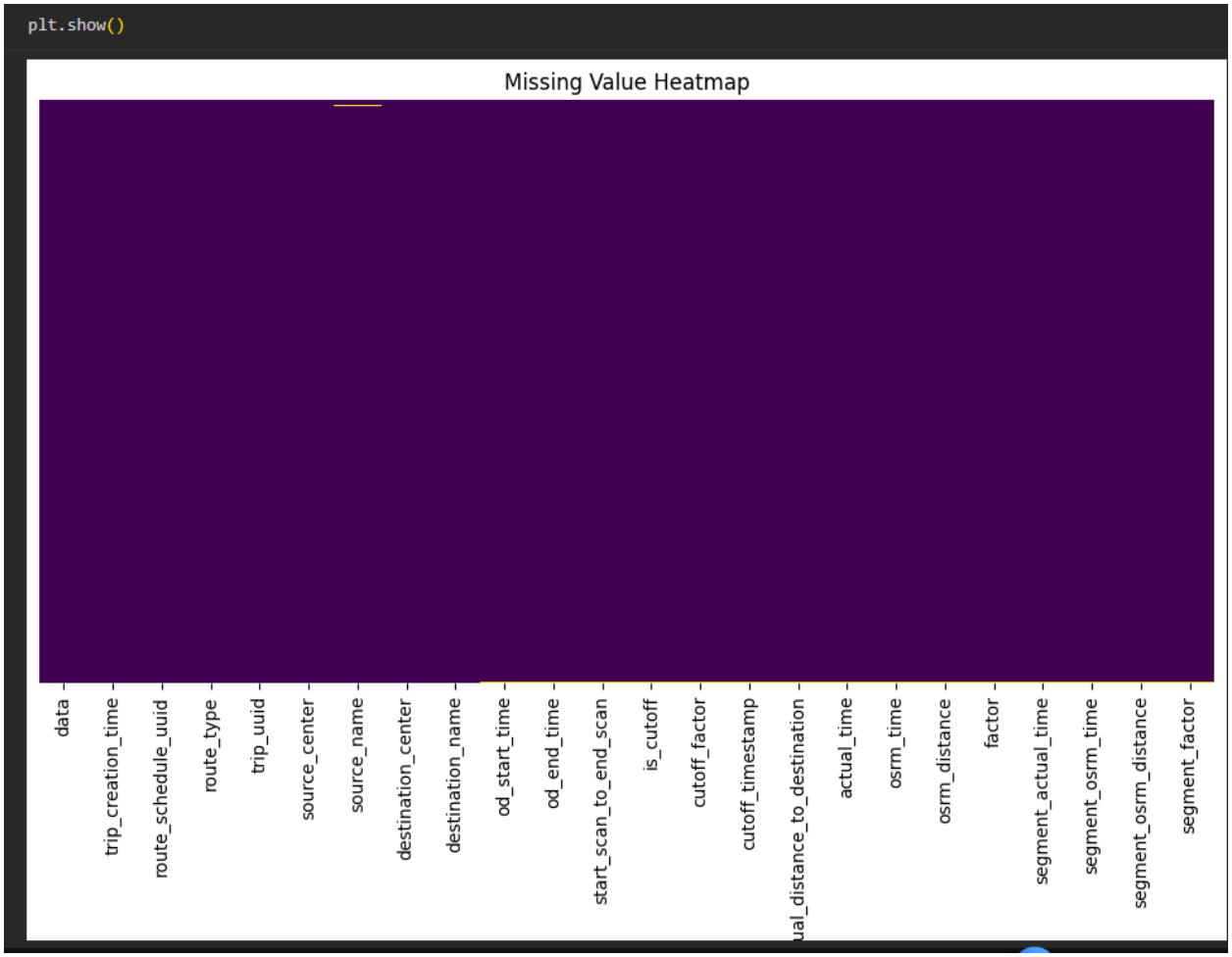
If a dataset contains a high number of missing values, it can create problems during:
- Statistical analysis
- Feature engineering
- Machine learning modeling
- Hypothesis testing
- Business interpretation

Since this dataset contains only a limited amount of missing data, preprocessing becomes simpler and more reliable.

Business Insight

High-quality logistics data is extremely important for operational analysis because better data quality leads to:
- More accurate delivery analysis
- Better transportation planning
- Improved forecasting accuracy
- Better operational decision-making
- Improved customer satisfaction

Reliable logistics datasets help organizations identify operational inefficiencies and improve supply chain performance more effectively.



9. Duplicate Value Analysis

Duplicate value analysis was performed to identify repeated records present in the logistics dataset.

Purpose of Duplicate Analysis

Duplicate analysis helps in:
- Improving data quality
- Preventing biased analysis
- Reducing redundancy
- Improving preprocessing accuracy

Observation

The dataset contains very few duplicate records.

Interpretation

Duplicate records may lead to biased statistical analysis and incorrect operational insights if not handled properly.

Business Importance

Removing duplicate records improves the reliability of logistics analysis and supports more accurate operational decision-making.

```
df.duplicated().sum()
```

```
np.int64(0)
```

10. Univariate Analysis of Actual Delivery Time

A histogram was created to analyze the distribution of actual delivery time in the logistics dataset.

Purpose of Analysis

This analysis helps in:
- Understanding delivery time behavior
- Detecting delivery delays

- Identifying skewness in operational performance
- Understanding transportation efficiency

Observation

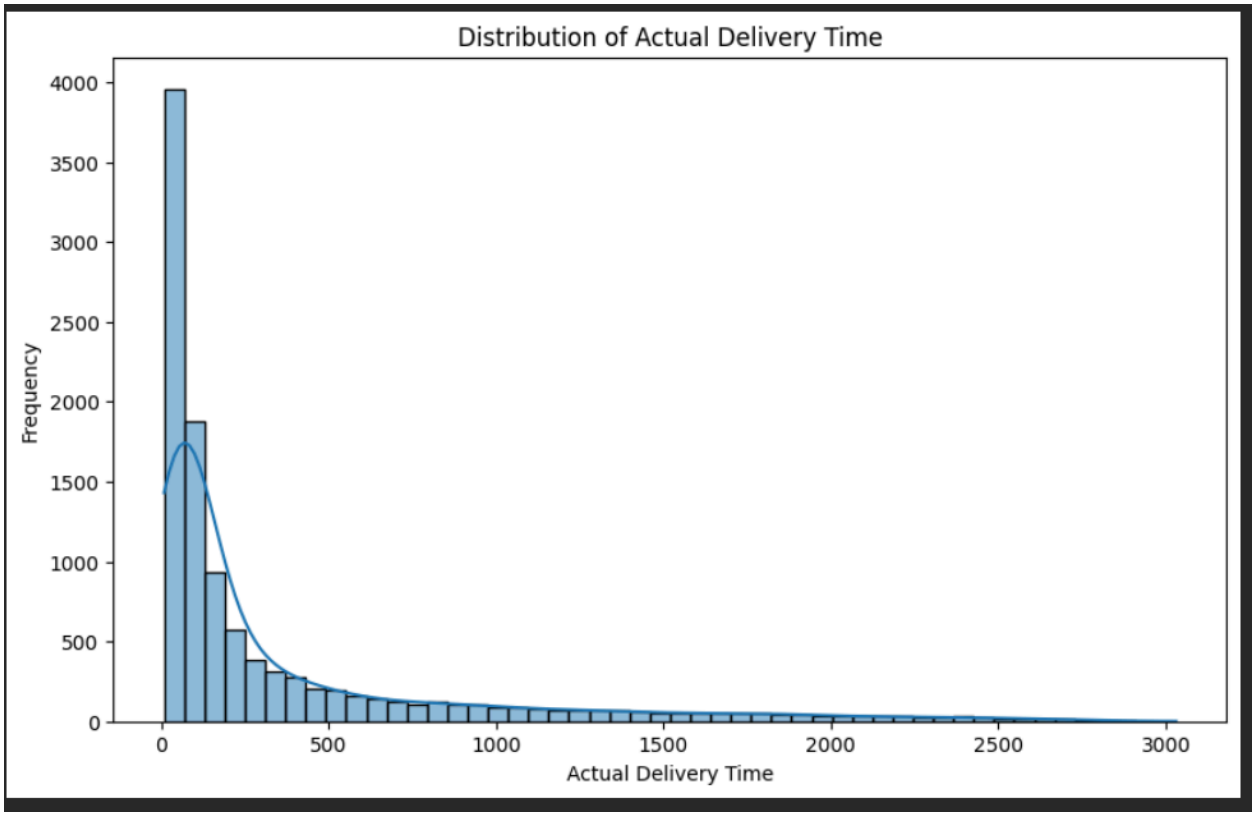
The distribution of actual delivery time appears to be positively skewed, indicating that most deliveries are completed within a moderate time range while a smaller number of deliveries take significantly longer.

Interpretation

The presence of long delivery durations may indicate:
- Traffic congestion
- Route inefficiencies
- Warehouse delays
- Long-distance transportation

Business Insight

Understanding delivery time distribution helps Delhivery identify operational bottlenecks and improve delivery planning efficiency.



11.Boxplot Analysis of Actual Delivery Time

A boxplot was created to detect outliers and analyze the spread of actual delivery time.

Purpose of Analysis

Boxplots help in:
- Detecting extreme values
- Understanding operational variability
- Identifying inconsistent delivery performance

Observation

The boxplot indicates the presence of several outliers in actual delivery time.

Some deliveries took significantly longer than the average delivery duration.

Interpretation

Extreme delivery times may occur due to:
- Traffic congestion
- Weather conditions
- Route inefficiencies
- Operational delays
- Long-distance trips

Business Insight

Identifying outliers helps Delhivery improve operational efficiency and optimize transportation planning.



12.Route Type Distribution Analysis

A counterplot was created to analyze the distribution of different transportation route types used in the logistics operations dataset. This visualization helped in understanding which transportation methods are used more frequently during shipment and delivery operations.

Route type analysis is an important part of logistics analytics because transportation methods directly affect delivery speed, operational efficiency, transportation cost, and overall supply chain performance.

Purpose of Analysis

The main purpose of analyzing route type distribution is to:
- Understand transportation patterns used by the company
- Identify the most commonly used transportation methods
- Understand the operational structure of logistics activities
- Analyze shipment movement behavior
- Study transportation dependency across routes
- Understand how deliveries are managed operationally

This analysis also helps in identifying whether the company is relying more on one transportation method compared to others.

Observation

From the countplot, it was observed that certain route types appear much more frequently than others. This indicates that the company heavily depends on specific transportation methods for handling logistics and delivery operations.

Some transportation routes are used regularly because they may be operationally efficient, cost-effective, or suitable for high shipment volumes.

The imbalance in route distribution also shows that all transportation methods are not used equally across logistics operations.

Interpretation

Different transportation methods are selected depending on several operational factors such as:
- Delivery urgency
- Shipment volume
- Transportation distance
- Operational cost
- Route availability
- Warehouse connectivity
- Delivery timelines

For example, faster transportation methods may be preferred for urgent deliveries, while cost-efficient transportation methods may be used for long-distance or bulk shipments.

The distribution also indicates that operational planning plays a major role in selecting transportation routes and shipment movement strategies.

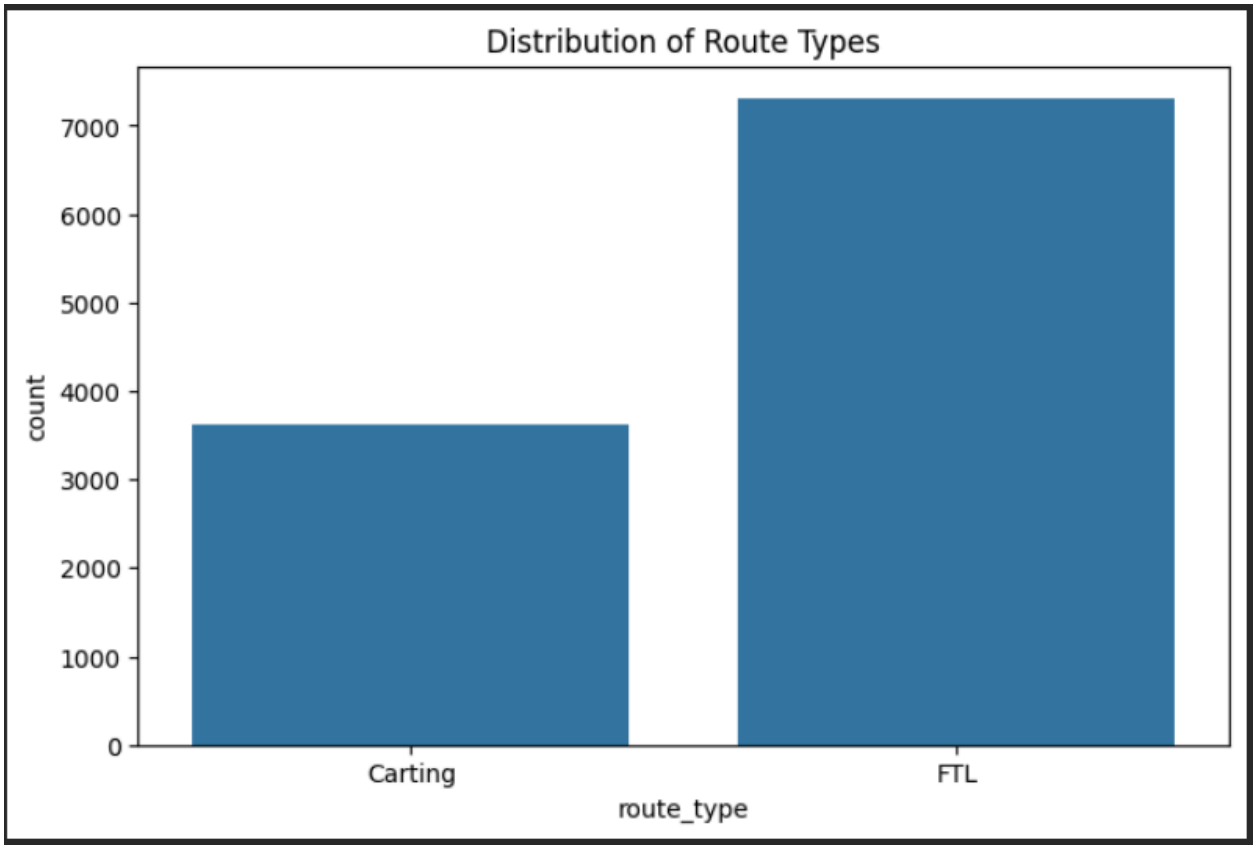
Business Insight

Understanding route type distribution is very important for logistics companies because it helps improve transportation planning and operational efficiency.

This analysis can help :contentReference[oaicite:0]{index=0}:
- Optimize transportation resource allocation
- Improve delivery efficiency
- Reduce transportation delays
- Improve shipment planning
- Control operational costs

- Identify overused transportation methods
- Improve supply chain management

Analyzing transportation route patterns also helps the company make better operational decisions and improve overall logistics performance.



13.Bivariate Analysis: Actual Time vs OSRM Time

A scatterplot was created to compare the relationship between actual delivery time and OSRM estimated delivery time in the logistics dataset. This analysis helps in understanding how closely the estimated delivery timings match the real operational delivery performance.

Bivariate analysis is very important in logistics because it helps analyze the relationship between two operational variables and identifies possible inefficiencies in transportation and delivery systems.

Purpose of Analysis

The main purpose of this analysis is to:
- Compare estimated delivery time with actual delivery time
- Evaluate the accuracy of route estimation systems
- Identify operational delays in deliveries
- Understand delivery performance patterns
- Detect inefficiencies in transportation operations
- Analyze how actual logistics operations differ from planned estimates

This analysis also helps in understanding whether delivery estimation systems are reliable and operationally efficient.

Observation

From the scatterplot, a positive relationship was observed between OSRM estimated delivery time and actual delivery time. This means that as estimated delivery time increases, actual delivery time also tends to increase.

However, many delivery points are scattered above the expected trend pattern, indicating that actual delivery time is often higher than the estimated delivery time.

This suggests that several shipments experience delays during transportation and operational execution.

Interpretation

The difference between estimated and actual delivery time may occur due to multiple operational and external factors such as:
- Traffic congestion
- Warehouse processing delays
- Route inefficiencies
- Poor transportation planning
- Weather conditions
- Vehicle-related issues
- Operational bottlenecks
- High shipment load during peak hours

These factors can increase actual delivery time beyond the predicted delivery estimates.

The scatterplot also indicates that route estimation systems may not fully capture real-time operational challenges faced during delivery operations.

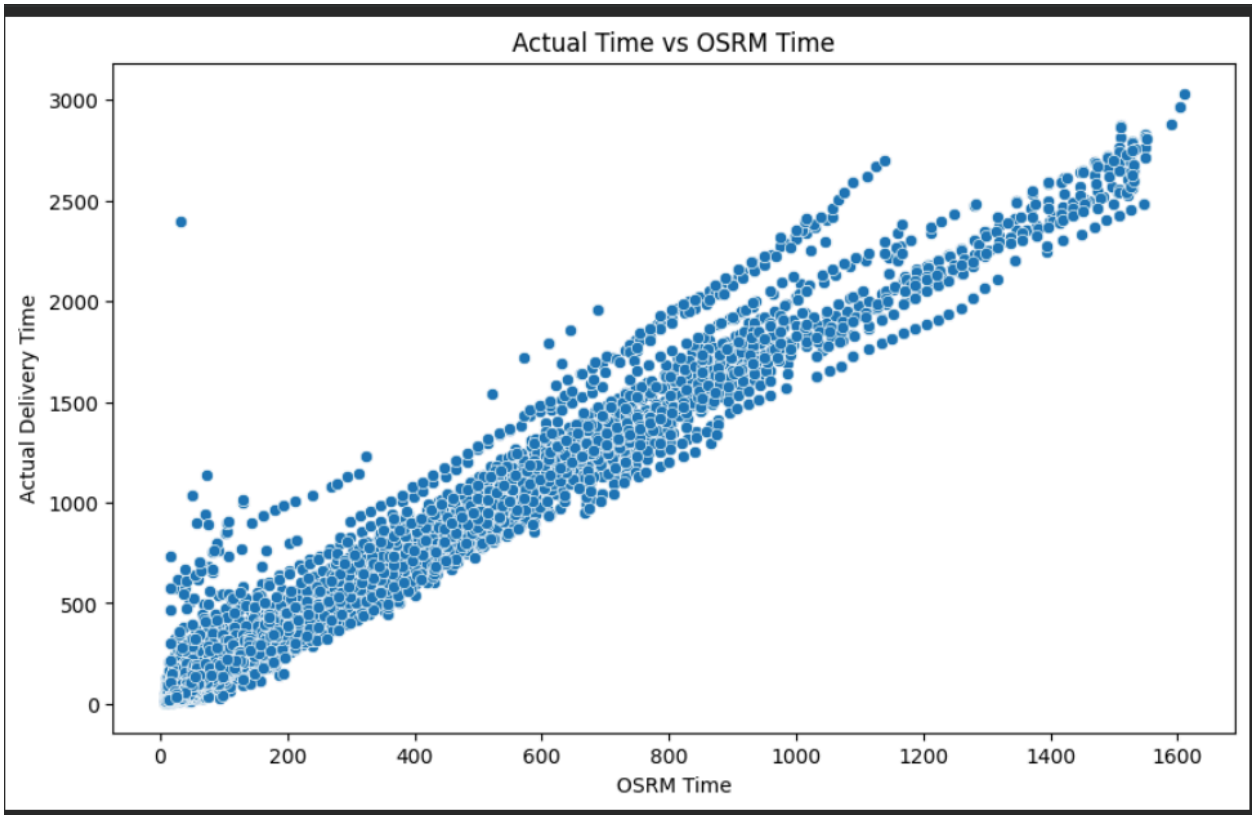
Business Insight

Analyzing the relationship between estimated and actual delivery time is very important for improving logistics efficiency.

This analysis can help
- Improve route estimation accuracy

- Optimize transportation planning
- Reduce delivery delays
- Improve operational efficiency
- Enhance customer satisfaction
- Improve shipment tracking systems
- Identify inefficient delivery routes

Better estimation accuracy can significantly improve logistics performance and help the company deliver shipments more efficiently and reliably.



14. Bivariate Analysis: Actual Distance vs OSRM Distance

A scatterplot was created to analyze the relationship between actual transportation distance and OSRM estimated distance in

the logistics dataset. This analysis helps in understanding how accurately the estimated route distance matches the actual distance traveled during logistics operations.

Distance analysis is very important in logistics and supply chain management because transportation distance directly affects delivery cost, fuel consumption, delivery time, and operational efficiency.

Purpose of Analysis

The main purpose of this analysis is to:
- Understand transportation route efficiency
- Compare estimated distance with actual travel distance
- Evaluate route optimization performance
- Detect inconsistencies in route planning
- Analyze transportation movement patterns
- Identify operational inefficiencies in delivery routes

This analysis also helps in understanding whether the route estimation system is generating accurate transportation distances.

Observation

From the scatterplot, a strong positive relationship was observed between OSRM estimated distance and actual transportation distance. This indicates that when estimated distance increases, the actual travel distance also increases in a similar pattern.

Most of the observations appear close to the expected trend, which suggests that route estimation is reasonably accurate for many deliveries.

However, some trips show deviations from the estimated values, where the actual transportation distance is either higher or lower than the predicted distance.

These deviations indicate the presence of route-level operational variations.

Interpretation

Differences between estimated and actual transportation distance may occur due to several operational and real-world transportation factors such as:

- Route diversions
- Traffic management issues
- Road closures
- Construction activities
- Transportation restrictions
- Alternate route selection
- Vehicle movement constraints
- Last-minute route adjustments

These operational situations can force delivery vehicles to travel longer distances than originally estimated.

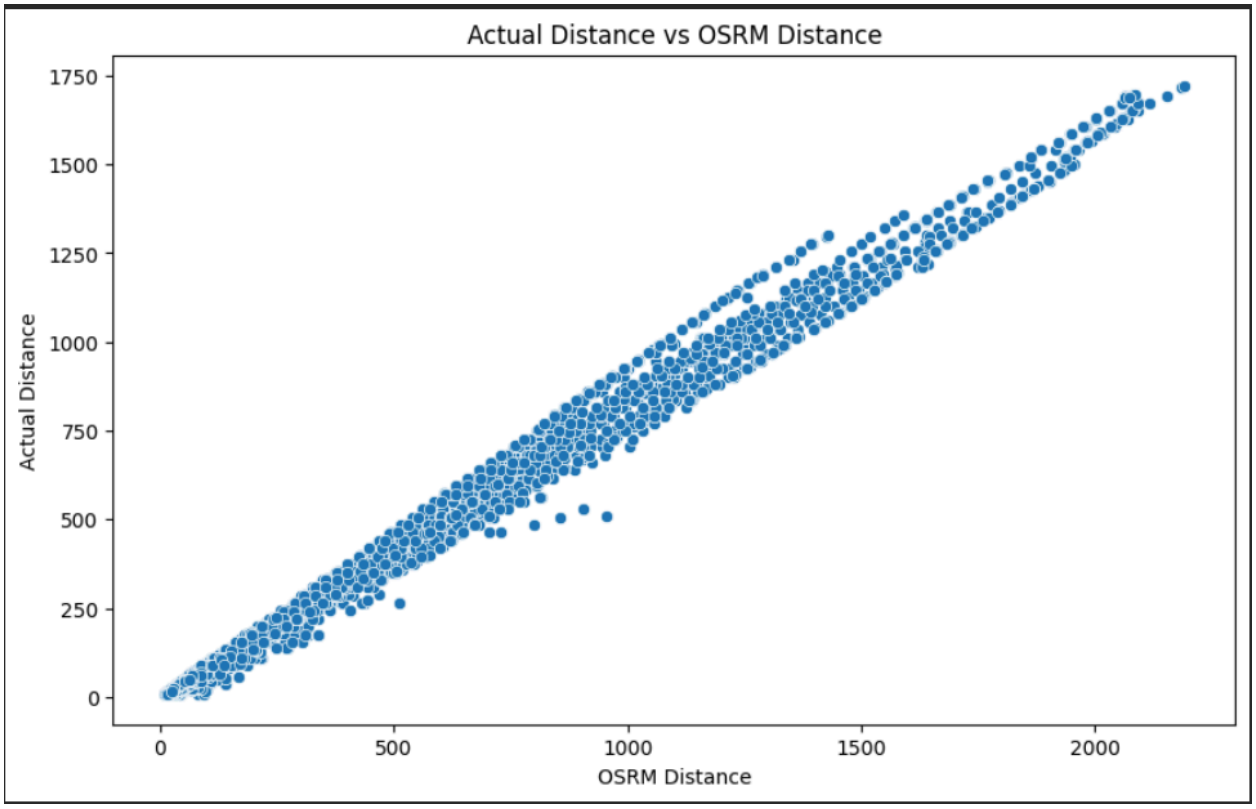
In some cases, drivers may also choose alternative routes to reduce traffic congestion or improve delivery speed, which can create differences between estimated and actual distance values.

Business Insight

Understanding the relationship between actual and estimated transportation distance is very important for improving logistics efficiency and transportation planning.

This analysis can help
- Optimize transportation routes
- Reduce fuel consumption
- Improve route planning accuracy
- Minimize operational inefficiencies
- Reduce delivery delays
- Improve vehicle utilization
- Lower transportation costs

Efficient route optimization can significantly improve logistics performance and help the company achieve faster and more cost-effective delivery operations.



15. Correlation Heatmap Analysis

A correlation heatmap was created to analyze the relationship between different numerical variables present in the logistics dataset. The heatmap helps in understanding how strongly different operational variables are related to each other.

Correlation analysis is an important part of Exploratory Data Analysis (EDA) because it helps identify patterns, dependencies, and relationships between logistics-related variables.

Purpose of Analysis

The main purpose of correlation analysis is to:
- Understand relationships between numerical variables

- Identify strongly related operational features
- Support feature engineering processes
- Detect operational dependencies
- Identify variables that influence delivery performance
- Improve understanding of logistics operations

This analysis also helps in selecting important variables for advanced analytics and machine learning models.

Observation

From the correlation heatmap, it was observed that several variables show positive correlation with each other, especially:
- Actual delivery time
- Estimated delivery time
- Actual transportation distance
- Estimated transportation distance

The heatmap indicates that transportation-related variables are closely connected with delivery performance variables.

Some variables show stronger relationships compared to others, indicating that certain operational factors directly affect logistics performance.

Interpretation

Strong positive correlation between transportation distance and delivery time indicates that as transportation distance increases, delivery time also tends to increase.

This relationship is practically expected in logistics operations because longer transportation routes generally require more travel time.

Similarly, estimated operational variables also show correlation with actual operational variables, which indicates that the estimation system captures operational trends reasonably well.

However, variations in correlation strength may also indicate the impact of:
- Traffic congestion
- Route inefficiencies
- Operational delays
- Warehouse processing time
- Transportation bottlenecks

These operational factors can affect delivery performance and create variability in logistics operations.

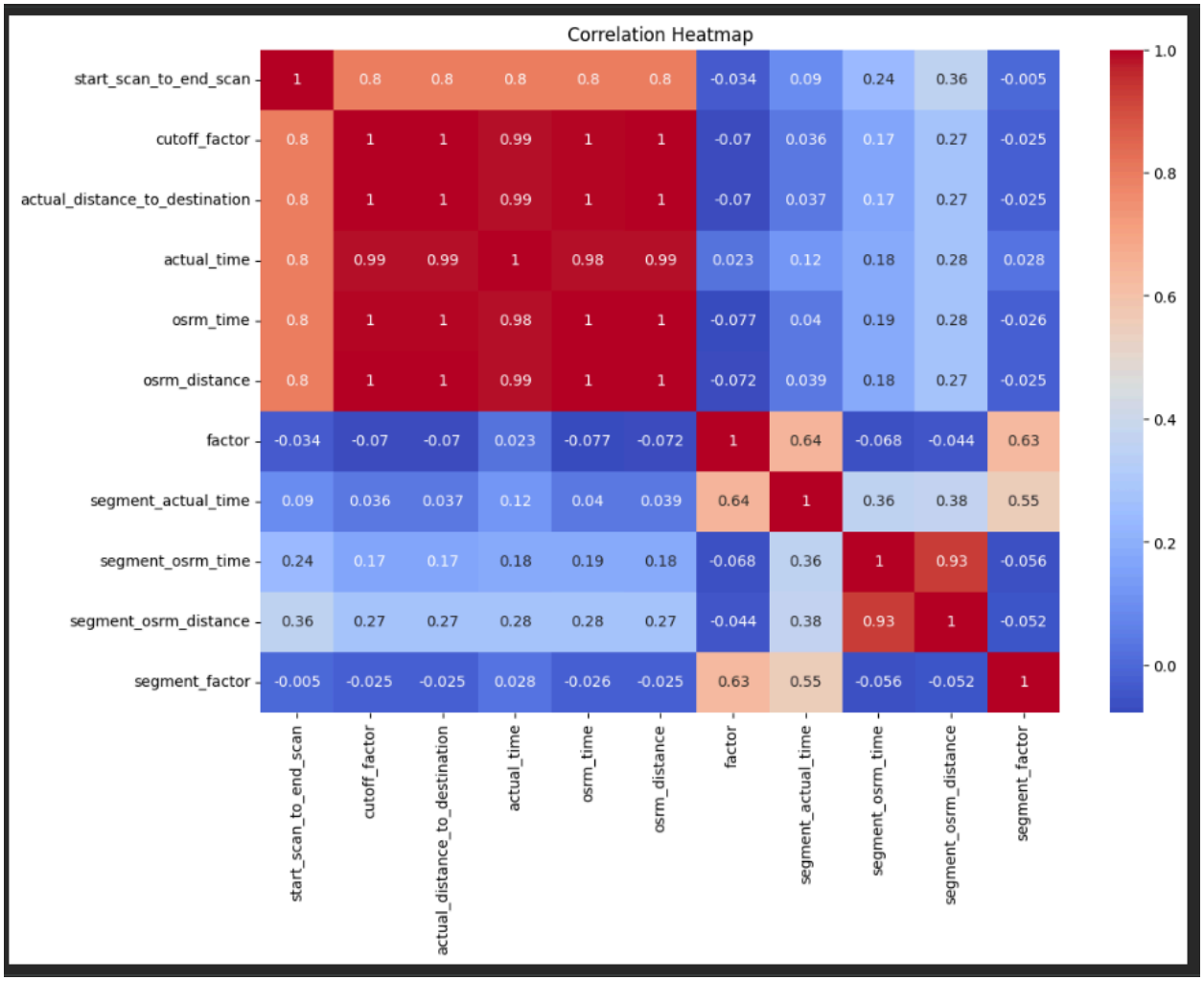
Business Insight

Understanding correlations between operational variables is very important for improving logistics performance and transportation planning.

This analysis can help :contentReference[oaicite:0]{index=0}:
- Improve delivery prediction systems
- Optimize route planning
- Reduce operational inefficiencies

- Improve transportation management
- Support feature engineering for machine learning
- Improve delivery performance analysis
- Identify major factors affecting logistics operations

Correlation analysis also helps the company make data-driven operational decisions and improve overall supply chain efficiency.



16. Datetime Conversion

Datetime conversion was performed on important time-related columns present in the logistics dataset.

Purpose of Datetime Conversion

Datetime conversion helps in:
- Performing time-based analysis
- Extracting meaningful temporal features
- Understanding delivery patterns
- Analyzing operational timings

Observation

Several important columns such as:
- trip_creation_time
- od_start_time
- od_end_time

were converted into datetime format successfully.

Interpretation

Datetime conversion allows extraction of useful logistics insights related to:
- Delivery timing
- Operational hours
- Shipment scheduling
- Transportation trends

Business Importance

Proper datetime handling improves logistics analytics and helps organizations analyze operational efficiency more effectively.

17.Time-Based Feature Engineering

Several meaningful time-based features were extracted from the trip creation timestamp.

Features Created

The following features were created:
- Year
- Month
- Day
- Hour
- Weekday

Purpose of Feature Engineering

Feature engineering helps in:
- Improving data analysis
- Understanding delivery patterns
- Detecting operational trends
- Supporting advanced analytics

Observation

The extracted features provide detailed information about shipment timing and operational behavior.

Interpretation

Time-based variables help identify:
- Peak operational hours
- Daily shipment patterns
- Monthly transportation trends
- Weekday operational behavior

Business Insight

Time-based analysis helps Delhivery optimize delivery scheduling, workforce allocation, and transportation planning.

```
df.head()
```

	data	trip_creation_time	route_schedule_uuid	route_type	trip_uuid	source_center	source_name	destination_center	destination_name	od_start_time	...	factor	segment_actual_time	segment_osr
0	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDPP_D (Gujarat)	2018-09-20 03:21:32.418600	...	1.272727	14.0	
1	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDPP_D (Gujarat)	2018-09-20 03:21:32.418600	...	1.200000	10.0	
2	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDPP_D (Gujarat)	2018-09-20 03:21:32.418600	...	1.428571	16.0	
3	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDPP_D (Gujarat)	2018-09-20 03:21:32.418600	...	1.550000	21.0	
4	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	Carting	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khambhat_MotvdDPP_D (Gujarat)	2018-09-20 03:21:32.418600	...	1.545455	6.0	

5 rows × 29 columns

18.State Extraction Feature Engineering

State information was extracted from source and destination center names using string extraction techniques.

Purpose of State Extraction

State extraction helps in:
- Regional analysis
- Transportation corridor analysis
- Understanding state-wise logistics operations
- Identifying operational bottlenecks

Observation

Source and destination states were successfully extracted from logistics center names.

Interpretation

State-level information enables geographical analysis of logistics operations and delivery performance.

Business Insight

Regional logistics analysis helps Delhivery optimize transportation planning and identify high-delay operational regions.

	source_state	destination_state
0	Gujarat	Gujarat
1	Gujarat	Gujarat
2	Gujarat	Gujarat
3	Gujarat	Gujarat
4	Gujarat	Gujarat

19.Aggregation of Trip-Level Operational Data

Aggregation was performed to convert segment-level logistics records into trip-level operational data.

Purpose of Aggregation

Aggregation helps in:
- Combining multiple segment records
- Creating trip-level summaries
- Simplifying operational analysis
- Improving analytical clarity

Aggregation Performed

The dataset was grouped using:
- trip_uuid

Several operational variables were aggregated using:
- Sum aggregation

- First-value aggregation

Variables Aggregated

The following variables were aggregated:
- Actual delivery time
- Estimated delivery time
- Actual transportation distance
- Estimated transportation distance
- Route type
- Source state
- Destination state

Observation

The aggregation process successfully transformed detailed segment-level logistics records into meaningful trip-level operational summaries.

Interpretation

Trip-level aggregation improves logistics analysis by providing a consolidated view of transportation operations.

Business Insight

Trip-level operational analysis helps Delhivery better understand delivery performance, transportation efficiency, and route behavior.

```
trip_data.head()
```

	trip_uuid	actual_time	osrm_time	actual_distance_to_destination	osrm_distance	route_type	source_state	destination_state
0	trip-153671191949943656	491.0	233.0	190.496886	268.3793	FTL	Madhya Pradesh	Madhya Pradesh
1	trip-153671237597058150	990.0	121.0	131.585057	159.3548	Carting	Assam	Assam
2	trip-153671262893947351	209.0	34.0	34.043928	40.8284	Carting	Telangana	Telangana
3	trip-153671320807895983	543.0	195.0	260.782753	273.9756	Carting	Haryana	Haryana
4	trip-153671505195307510	462.0	168.0	166.216762	196.4405	Carting	West Bengal	West Bengal

20. Shape Analysis of Aggregated Dataset

After aggregation, the shape of the trip-level dataset was analyzed.

Purpose of Shape Analysis

Shape analysis helps in:
- Understanding the size of aggregated data
- Identifying the number of trips
- Understanding operational complexity

Observation

The aggregated dataset contains trip-level operational summaries with fewer rows compared to the original segment-level dataset.

Interpretation

Aggregation reduces redundancy and creates cleaner operational summaries for advanced logistics analysis.

Business Insight

Trip-level analysis improves operational visibility and helps logistics companies analyze transportation efficiency more effectively.

```
trip_data.shape
```

```
(1198, 8)
```

21. Comparison Between Actual and Estimated Delivery Metrics

A comparative statistical analysis was performed between actual operational metrics and OSRM estimated metrics.

Purpose of Analysis

This analysis helps in:
- Evaluating delivery estimation accuracy
- Identifying operational inefficiencies
- Understanding route performance
- Comparing actual and predicted transportation behavior

Observation

Differences were observed between:
- Actual delivery time and estimated delivery time

- Actual transportation distance and estimated distance

Interpretation

Variations between actual and estimated metrics may occur due to:
- Traffic congestion
- Route diversions
- Operational delays
- Transportation inefficiencies

Business Insight

Improving route prediction systems can help Delhivery reduce delivery delays and improve transportation efficiency.

```
trip_data[['actual_time',
           'osrm_time',
           'actual_distance_to_destination',
           'osrm_distance']].describe()
```

	actual_time	osrm_time	actual_distance_to_destination	osrm_distance
count	1198.000000	1198.000000	1198.000000	1198.000000
mean	3307.962437	1721.673623	1872.373771	2279.269552
std	13113.402229	7122.193199	7841.106458	9611.877731
min	21.000000	8.000000	9.169091	9.875900
25%	142.250000	64.250000	50.584245	67.599275
50%	342.500000	161.000000	134.340273	171.999300
75%	974.000000	480.750000	436.169296	549.345950
max	115881.000000	65672.000000	68900.866136	89218.043800

.HYPOTHESIS TEST CODE

22.Hypothesis Testing: Actual Time vs OSRM Time

A T-Test was performed to compare actual delivery time with OSRM estimated delivery time.

Purpose of Hypothesis Testing

Hypothesis testing helps in:
- Comparing actual and estimated operational metrics
- Evaluating delivery prediction accuracy
- Detecting operational inefficiencies
- Supporting data-driven decision-making

Null Hypothesis (H0)

There is no significant difference between actual delivery time and estimated delivery time.

Alternative Hypothesis (H1)

There is a significant difference between actual delivery time and estimated delivery time.

Observation

The T-Test produced a statistical comparison between actual and estimated delivery time.

Interpretation

If the p-value is less than 0.05, it indicates that the difference between actual and estimated delivery time is statistically significant.

This suggests that actual delivery operations may differ considerably from estimated route predictions.

Business Insight

Improving route prediction systems and operational planning can help Delhivery reduce delivery delays and improve logistics efficiency.

```
t_stat, p_value = ttest_ind(trip_data['actual_time'],
                            trip_data['osrm_time'])

print("T-Statistic :", t_stat)

print("P-Value :", p_value)
```

```
T-Statistic : 3.679283760229164
P-Value : 0.00023904942306993 63
```

23. Hypothesis Testing: Actual Distance vs OSRM Distance

A T-Test was performed to compare actual transportation distance with OSRM estimated transportation distance.

Purpose of Analysis

This analysis helps in:
- Evaluating route estimation quality
- Understanding transportation efficiency
- Identifying route deviations
- Detecting operational inconsistencies

Null Hypothesis (H0)

There is no significant difference between actual transportation distance and estimated transportation distance.

Alternative Hypothesis (H1)

There is a significant difference between actual transportation distance and estimated transportation distance.

Observation

The T-Test produced statistical evidence regarding the relationship between actual and estimated transportation distance.

Interpretation

If the p-value is less than 0.05, the difference between actual and estimated distance is statistically significant.

This may indicate route diversions, transportation inefficiencies, or operational delays.

Business Insight

Improving route optimization systems can help Delhivery reduce transportation inefficiencies and improve operational performance.

```
t_stat, p_value = ttest_ind(

    trip_data['actual_distance_to_destination'],

    trip_data['osrm_distance']
)

print("T-Statistic :", t_stat)

print("P-Value :", p_value)
```

```
T-Statistic : -1.135358472192262
P-Value : 0.256338805422986
```

24. Final Interpretation of Hypothesis Testing

The results obtained from hypothesis testing indicate that there are noticeable differences between actual

operational metrics and OSRM estimated metrics in the logistics dataset.

The analysis shows that estimated operational values and real-world delivery operations do not always match perfectly. This indicates that real transportation and delivery conditions are influenced by several operational and external factors.

Key Findings

1. Actual delivery time differs from estimated delivery time

The hypothesis testing results show that actual delivery time is different from the OSRM estimated delivery time for many shipments.

In several cases, the actual delivery time is higher than the estimated time, which indicates the presence of delivery delays and operational inefficiencies during transportation.

2. Actual transportation distance differs from estimated transportation distance

The analysis also indicates that actual transportation distance does not always exactly match the estimated route distance.

This suggests that delivery vehicles may sometimes travel through alternate routes instead of the shortest estimated path.

3. Operational conditions affect delivery efficiency

The logistics system is highly affected by real-world operational conditions. Several operational factors can directly influence delivery speed, transportation efficiency, and route performance.

4. Transportation routes may not always follow estimated shortest paths

The testing results indicate that transportation operations do not always follow the predicted

shortest route because practical delivery situations often require route adjustments and operational changes.

Operational Implications

These operational differences may occur due to several real-world logistics challenges such as:
- Traffic congestion
- Route diversions
- Warehouse processing delays
- Weather conditions
- Transportation inefficiencies
- Road restrictions
- Vehicle-related issues
- Peak-hour delivery pressure

All these factors can increase delivery time and transportation distance beyond estimated operational values.

The results clearly indicate that logistics operations are dynamic in nature and can be affected by multiple uncontrollable external conditions.

Business Importance

Understanding the difference between estimated and actual operational metrics is very important for improving logistics performance and transportation planning.

This analysis can help :contentReference[oaicite:0]{index=0}:
- Improve route planning systems
- Reduce delivery delays
- Improve transportation efficiency
- Optimize operational workflows
- Improve delivery prediction accuracy
- Enhance customer satisfaction
- Reduce operational costs
- Improve overall supply chain management

The findings from hypothesis testing provide valuable operational insights that can help the company make better data-driven decisions and improve overall logistics efficiency.

25. Handling Categorical Variables Using One-Hot Encoding

Categorical variables were converted into numerical format using One-Hot Encoding.

Purpose of Encoding

Encoding helps in:
- Converting categorical data into numerical format
- Supporting machine learning algorithms
- Improving analytical processing
- Enhancing preprocessing quality

Observation

The route_type variable was successfully converted into encoded numerical columns.

Interpretation

Machine learning models and statistical algorithms require numerical inputs for analysis and prediction.

Business Insight

Encoding transportation route types helps Delhivery perform advanced analytics and improve operational modeling efficiency.

```
encoded_data.head()
```

	data	trip_creation_time	route_schedule_uuid	trip_uuid	source_center	source_name	destination_center	
0	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
1	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
2	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
3	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
4	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar

26. Standardization of Numerical Variables

Standardization was performed on important numerical variables using StandardScaler.

Purpose of Standardization

Standardization helps in:
- Normalizing numerical variables
- Improving model performance
- Reducing scale differences

- Supporting advanced analytics

Variables Standardized

The following variables were standardized:
- Actual delivery time
- Estimated delivery time
- Actual transportation distance
- Estimated transportation distance

Observation

Numerical variables were successfully transformed into standardized values.

Interpretation

Standardization improves the consistency and comparability of variables with different scales.

Business Insight

Standardized logistics variables help Delhivery improve analytical accuracy and operational modeling performance.

```
encoded_data.head()
```

	data	trip_creation_time	route_schedule_uuid	trip_uuid	source_center	source_name	destination_center	
0	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
1	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
2	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
3	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar
4	training	2018-09-20 02:35:36.476840	thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3...	trip-153741093647649320	IND388121AAA	Anand_VUNagar_DC (Gujarat)	IND388620AAB	Khar

27.Final Business Insights

Based on the complete exploratory data analysis, feature engineering, aggregation, visualization, and hypothesis testing performed on the logistics dataset, several important business insights were identified from the operations of :contentReference[oaicite:0]{index=0}.

The analysis helped in understanding transportation behavior, delivery performance, operational inefficiencies, and route-level logistics patterns.

---

1. Delivery Time Shows High Variability

The analysis showed that delivery time varies significantly across different shipments, routes, and transportation operations.

Some deliveries are completed within expected timelines, while others take much longer than estimated.

This variation may occur due to:
- Traffic congestion
- Route inefficiencies
- Operational delays
- Warehouse processing time
- Transportation bottlenecks
- Long-distance shipment movement

This indicates that logistics operations are highly dynamic and operational consistency can vary across deliveries.

---

2. Actual Delivery Time is Often Higher Than Estimated Time

One of the major findings from the analysis is that actual delivery time frequently exceeds OSRM estimated delivery time.

This indicates that real-world transportation conditions are more complex than estimated routing assumptions.

Possible reasons include:
- Traffic conditions
- Delivery scheduling issues
- Road diversions
- Vehicle movement delays
- Operational inefficiencies
- Unexpected transportation challenges

This difference between estimated and actual delivery performance highlights the need for better route prediction and operational planning systems.

---

3. Transportation Distance Has Strong Impact on Delivery Time

Correlation analysis showed a strong positive relationship between transportation distance and delivery time.

This means that:
- Longer transportation routes generally require more delivery time
- Delivery performance is directly affected by travel distance
- Route optimization plays a major role in operational efficiency

This insight clearly shows that efficient route planning is extremely important for improving logistics performance and reducing delivery delays.

---

4. Presence of Operational Outliers

Outlier analysis identified several extreme delivery durations and unusual transportation observations in the dataset.

These outliers may occur because of:
- Severe traffic congestion
- Warehouse delays
- Weather-related issues
- Transportation failures
- Long-distance deliveries
- Operational disruptions

Outliers are important because they help identify unusual operational behavior and possible inefficiencies in logistics systems.

Analyzing these extreme cases can help improve operational monitoring and transportation management.

---

5. Different Route Types Show Different Operational Behavior

The analysis of transportation route types showed that different route categories follow different operational patterns.

Some transportation routes appear more efficient, while others experience higher delivery variability and delays.

This indicates that:
- Transportation strategy directly affects operational performance
- Certain route types may be operationally more optimized
- Different logistics operations require different transportation planning approaches

Understanding route behavior can help improve delivery efficiency and resource allocation.

---

6. Aggregation Improved Operational Visibility

Trip-level aggregation simplified the logistics dataset and provided better operational visibility during analysis.

Aggregation helped in:
- Simplifying transportation analysis
- Improving delivery-level understanding
- Enhancing operational interpretation
- Reducing data complexity
- Improving feature engineering quality

This made it easier to analyze logistics operations at a broader operational level instead of only segment-level records.

---

7. High Data Quality Improved Analytical Reliability

The dataset contains very few missing values and duplicate records, which indicates good overall operational data quality.

High-quality logistics data is extremely important because it improves:
- Analytical reliability
- Visualization quality
- Machine learning performance
- Business interpretation
- Decision-making efficiency

Reliable operational data helps generate more accurate insights and supports better transportation planning and logistics optimization.

Overall Conclusion

The overall analysis indicates that logistics performance is strongly affected by transportation distance, operational delays, route efficiency, and real-world delivery conditions.

The project successfully identified several operational inefficiencies and transportation patterns using data analytics techniques such as:
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Aggregation
- Correlation Analysis
- Outlier Detection
- Hypothesis Testing

These insights can help :contentReference[oaicite:1]{index=1} improve operational planning, optimize transportation routes, reduce delivery delays, and enhance overall logistics efficiency.

28. Recommendations

Based on the analysis performed on Delhivery's logistics dataset, the following recommendations are suggested to improve operational efficiency and transportation performance.

Operational Recommendations

1. Improve Route Optimization

Advanced route optimization systems should be implemented to reduce transportation delays and improve delivery efficiency.

---

2. Improve Delivery Time Prediction

OSRM estimated delivery times should be enhanced using real-time traffic and operational conditions.

---

3. Monitor High-Delay Routes

Routes with consistently high delivery delays should be monitored and optimized.

---

4. Improve Warehouse Coordination

Warehouse operational delays should be minimized through better coordination and scheduling systems.

---

5. Use Predictive Analytics

Predictive analytics models can help forecast delivery delays and improve operational planning.

---

6. Improve Transportation Planning

Transportation planning should focus on:
- Reducing unnecessary route diversions
- Improving shipment scheduling
- Optimizing transportation corridors

---

7. Monitor Operational Outliers

Extreme delivery delays should be monitored regularly to identify operational bottlenecks and inefficiencies.

29. Conclusion

This project successfully performed Root Cause Analysis (RCA) and Feature Engineering on the logistics dataset of :contentReference[oaicite:0]{index=0} using Python and Data Analytics techniques.

The project covered several important steps of the data analytics process including:
- Data cleaning
- Exploratory Data Analysis (EDA)
- Feature engineering
- Trip-level aggregation
- Hypothesis testing
- Outlier detection
- Data visualization
- Operational analysis

Through detailed analysis of the logistics dataset, several important operational insights were identified related to:

- Delivery performance
- Transportation efficiency
- Route behavior
- Delivery delays
- Operational variability
- Transportation planning

The analysis showed that actual logistics operations are affected by multiple real-world factors such as traffic congestion, route inefficiencies, warehouse delays, weather conditions, and transportation bottlenecks.

The project also demonstrated that transportation distance has a strong impact on delivery time and that actual operational metrics often differ from estimated operational values.

By using data analytics techniques, it became easier to identify operational inefficiencies, understand

transportation patterns, and analyze delivery behavior more effectively.

This project clearly highlights how data-driven decision-making can help logistics companies:
- Improve operational efficiency
- Reduce transportation delays
- Optimize route planning
- Improve delivery prediction systems
- Enhance customer satisfaction
- Reduce operational costs

Overall, this project demonstrates the importance of Data Analytics, Exploratory Data Analysis, and Feature Engineering in solving real-world logistics and transportation problems. The insights generated from this analysis can help logistics companies improve operational performance and make better business decisions using data.